

Kathrah Platform: Technical Proposal & Implementation Guide

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Project: Kathrah - Muslim Intellectual Capital Database & Mentorship Hub

1. Project Overview

Kathrah is a visionary platform designed to preserve, connect, and mobilize Muslim intellectual capital through a centralized digital database and structured mentorship ecosystem. This document outlines the technical roadmap to build and launch the platform, including system architecture, technology stack, hosting strategy, security, email services, payment gateways, cost projections, marketing strategy, and post-launch agreement terms.

2. Platform Architecture

Core Stack:

- **Backend:** Django (Python)
- **Frontend:** React.js
- **Database:** PostgreSQL
- **Hosting:** AWS (Lightsail, RDS, S3, SES)
- **Authentication:** Google & Email (with JWT)
- **File Storage:** AWS S3 (PDFs, profile images)
- **Payment Gateway:** PayFast
- **Email Services:** Amazon SES or Mailgun

Modular Components:

1. User Management

- Sign up/login (email & Google)
- Roles: Admin, Mentor, Mentee

2. Intellectual Capital Database

- Dynamic profile creation
- Searchable by field, sector, expertise

3. Mentorship Hub

- Mentor application & approval
- Mentee requests, chat, scheduling (Google Meet integration)

4. Admin Panel

- Manage users, mentorship matches, reports

5. Subscription/Payments

- Membership payment via PayFast
- Invoice tracking

6. Content & Document Uploads

- Hosted on S3 with access control

3. Cost Estimate (AWS)

Year 1 - AWS Free Tier:

- Lightsail Instance (first 3 months free)
- RDS (750 hours/month + 20GB storage) — Free

- S3 (5GB storage) — Free
- SES (62,000 emails/month) — Free if sent from Lightsail

After Year 1 (Est. Monthly Cost):

Service	Lightsail (1GB RAM)	EC2 (t3.micro)	Notes
Hosting	~\$5/month	~\$8–\$15/month	Lightsail simpler; EC2 more flexible/scalable
Database (RDS)	~\$13–\$15	~\$13–\$15	PostgreSQL (db.t3.micro)
S3 Storage	~\$0.50	~\$0.50	20GB data
SES (Emails)	~\$0.50	~\$0.50	~5,000 emails
Backups & Misc.	~\$5–10	~\$5–10	Snapshots, bandwidth, etc.

Total Estimation After Year 1:

- **Lightsail Setup:** ~\$25–30/month (R450–R550/month)
- **EC2 Setup:** ~\$30–40/month (R550–R750/month)

SQLite can be used for development and testing, but **PostgreSQL (via RDS)** is strongly recommended for production to support scaling, concurrency, and data integrity.

4. Payment Gateway: PayFast

- **Setup Fee:** Free
- **Transaction Fee:**
 - 3.5% per transaction (Credit/Debit card, Instant EFT)

- No monthly fee
 - **Features:** PCI-compliant, secure checkout, mobile friendly
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5. Design Process & UI Mockups

Sources for design Inspiration:

- [Landingfolio](#)
- [Mobbin](#)
- [UI8](#)
- [Dribbble](#)

Process:

1. Create React UI mockups based on:
 - Landing Page
 - Profile Directory
 - Mentorship Matching
 - Admin Dashboard
 2. Provide client with 4–6 mockup themes for final selection
 3. Finalize UI direction and share Figma file or links
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6. Security Implementation

- **Django Security Features:**

- CSRF protection
 - XSS protection
 - Encrypted passwords (PBKDF2)
 - Secure headers and middleware
 - **Additional:**
 - HTTPS (via SSL Certificate)
 - Two-factor authentication (optional)
 - Email verification upon signup
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7. Deployment Strategy: React + Django on AWS

Approach 1: Single Lightsail Instance (Simpler)

- Use NGINX to reverse proxy:
 - Serve React build files
 - Proxy API requests to Django backend
- Host both frontend and backend on same Lightsail instance

Approach 2: Separate Hosting (Scalable)

- Frontend hosted on S3 + CloudFront (static site)
 - Django API hosted via Lightsail or EC2
 - CORS configured on backend
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8. Email Notifications

- **Amazon SES** (preferred)
 - Cheap, native to AWS
 - Verified domains
 - Deliverability support
 - R15–R30/month depending on volume
- **Alternative: Mailgun**
 - Free up to 5,000 emails/month
 - Easy setup, analytics dashboard

Use cases:

- Signup verification
- Mentorship status updates
- Admin alerts

9. Marketing Strategy

To maximize outreach and user adoption, the following marketing strategies will be implemented:

- **Radio Advertising:** Ads on community-focused stations such as Radio 786, Voice of the Cape, and CII.
 - **Influencer Marketing:** Partnering with influencers on Facebook, TikTok, and Instagram to spread awareness.
 - **University Outreach:** Engaging with MSA (Muslim Students' Association) chapters across universities to onboard young professionals and students.
 - **Targeted Digital Ads:** Running Facebook and Instagram ads targeting relevant demographics.
 - **Community Events:** Sponsoring and attending Islamic conferences and networking events.
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10. Final Agreement & Payment Terms

Total Development Fee: [REDACTED],000

Payment Terms:

- [REDACTED],000 to be paid upon project completion and handover
- Project will include:
 - Full source code (GitHub or private repo)
 - Documentation for deployment
 - Live hosted platform
 - Admin and user credentials

Maintenance & Support:

- **Free Maintenance:** 5 months post-launch
 - Covers bug fixes, minor changes
- **New Features:** R100/hour (as quoted)

Optional Partnership Proposal:

- 10% equity in Kathrah platform in lieu of cash payment
 - In exchange for long-term ongoing maintenance and updates and adding new features or major changes.

Please confirm if you accept this agreement and proposal so we can proceed with the next steps.

11. Next Steps

- Review mockups and select preferred UI design
- Approve project scope and terms
- Sign off to begin development phase
- First demo ready within 2–4 weeks

Thank you for trusting me with this meaningful project. I am excited to contribute to something that can have a lasting impact on the Muslim community in South Africa and beyond.

Abdurahman
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